

INDIAN STATISTICAL INSTITUTE

Analysis on Lie groups
Final Examination: 2025-26
M.Math 2nd Year
Date: 06.05.2026

Maximum Marks: 50

Maximum Time: 3 hrs.

[Answer as many questions as you can but maximum you can score is 50].

✓(1) Let G be a compact group, and $f \in C(G)$ be such that

$$\sum_{\delta \in \widehat{G}} d_{\delta}^{3/2} \|\widehat{f}(\delta)\|_{HS} < \infty.$$

Show that the Fourier series of f converges absolutely and uniformly to f . In particular, for $G = SU(2)$ and $f \in C^2(G)$, show that the Fourier series of f converges absolutely and uniformly to f . [10]

✓(2) Let G be a compact group, and μ the normalised Haar measure on G .

(a) Let $\psi \in C(G)$. Show that $\psi = d_{\pi}^{-1} \chi_{\pi}$ for some irreducible representation π of G if and only if ψ satisfies

$$\int_G \psi(ghg^{-1}k) dg = \psi(h)\psi(k), \quad \forall h, k \in G.$$

(b) Let $f \in L^2(G)$ be a non-trivial function. Show that

$$\mu(\text{supp}(f)) \left(\sum_{\delta \in \text{supp}(\widehat{f})} d_{\delta}^2 \right) \geq C,$$

where d_{δ} is the dimension of the space where δ is realised.

[5+5=10]

- (3) (a) Show that for every continuous class function $f : SU(2) \rightarrow \mathbb{C}$

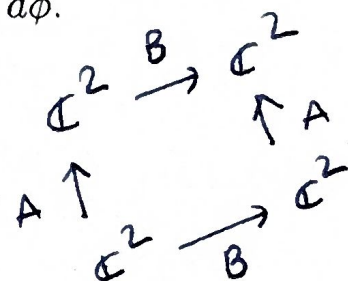
$$\int_{SU(2)} f(g) d\mu(g) = \int_0^{2\pi} f(t_\phi) \frac{\sin^2 \phi}{\pi} d\phi$$

where μ is the normalised Haar measure on $SU(2)$. Using this or otherwise show that for $f \in C(SU(2))$ one has

$$\int_{SU(2)} f(g) d\mu(g) = \int_0^{2\pi} \int_{SU(2)} f(gt_\phi g^{-1}) \frac{\sin^2 \phi}{\pi} d\mu(g) d\phi.$$

Here

$$t_\phi := \begin{pmatrix} e^{i\phi} & 0 \\ 0 & e^{-i\phi} \end{pmatrix} \quad (\phi \in \mathbb{R}).$$



- (b) Using (a), for $n \in \mathbb{N}$, show that

$$\int_{SU(2)} \chi_n(g) d\mu(g) = 0,$$

where χ_n is the character of the irreducible representation π_n (as defined in the class!)

[7+3=10]

- (4) (a) State the one parameter subgroup theorem.

- (b) Show that the center $Z(G)$ of a connected linear Lie group G is a closed subgroup of G . Prove that the Lie algebra of $Z(G)$ is given by $\ker(ad)$.

- (c) Find the Lie algebra of $SO(3)$.

[2+3+5=10]

- (5) (a) Show that the exponential map $\exp : \text{Lie}(SU(2)) \rightarrow SU(2)$ is surjective.

- (b) Show that the exponential map $\exp : \text{Lie}(SL(2, \mathbb{R})) \rightarrow SL(2, \mathbb{R})$ is not surjective by showing that the range is equal to

$$\{A \in SL(2, \mathbb{R}) : \text{Tr}(A) > -2\} \cup \{-I\}.$$

$$\begin{pmatrix} -2 & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$$

[5+5=10]

- (6) Let G be a compact group, and $\delta, \rho \in \widehat{G}$ be inequivalent representations.

- (a) Show that $f * h = 0$ for any $f \in C(G)_\delta$, and $h \in C(G)_\rho$.

- (b) Prove that $\chi_\delta * \chi_\delta = \frac{1}{d_\delta} \chi_\delta$.

- (c) Prove that the orthogonal projection $E_\delta : L^2(G) \rightarrow C(G)_\delta$ can be written as

$$E_\delta f = d_\delta f * \chi_\delta \quad (f \in L^2(G)).$$

[10]

- (7) (a) Let $\lambda \in \mathbb{C} \setminus \{-n(n+2) : n \in \mathbb{N}\}$, and $f \in C(SU(2))$. Using Fourier analysis find an expression for the unique solution of the following PDE:

$$-\Delta_G u + \lambda u = f.$$

- (b) For $n = 2, 3$, determine whether the natural representation of $SO(n)$ on $\mathbb{C}^n$ is irreducible.

[5+5=10]

$$Z \mapsto A Z$$